Answer **all** the questions in the spaces provided.

1 Fig. 1.1. shows an airplane dropping a crate of emergency supplies.

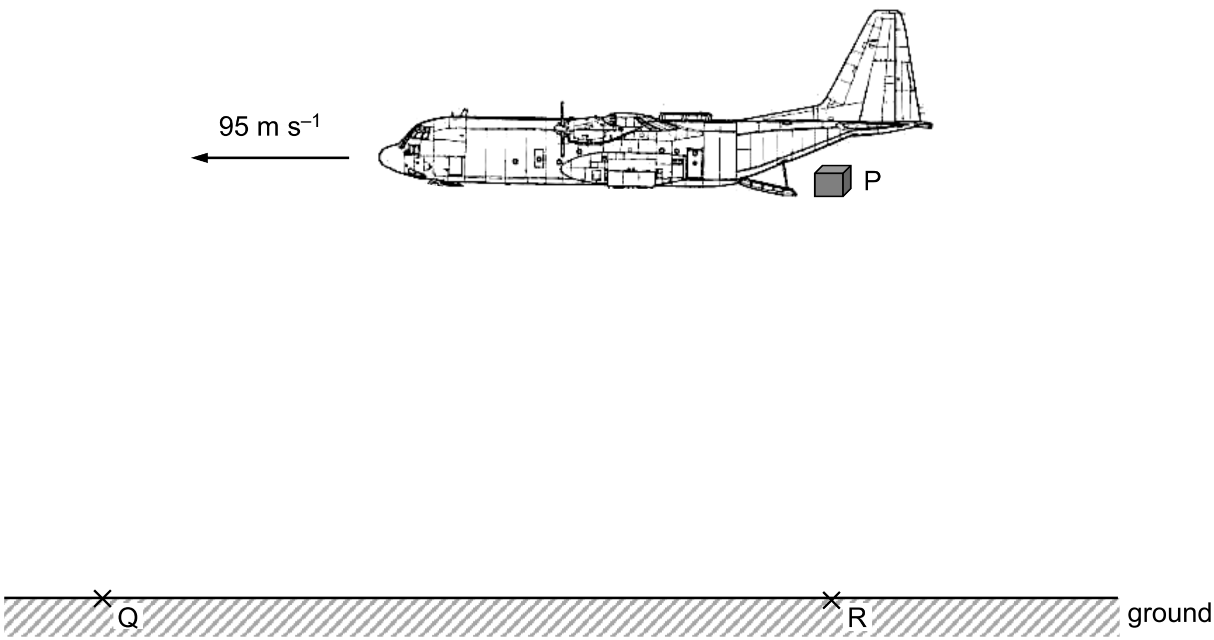

Fig. 1.1

The airplane is travelling horizontally at a constant speed 95 m s^{-1}. The crate is released at point P and it lands at point Q.

The air resistance acting on the crate may be neglected.

(a) On Fig. 1.1, sketch the path of the crate. [1]

(b) Explain why the horizontal component of the crate's velocity is constant when it is moving between point P and Q.

....

.... [1]

(c) The maximum vertical component of the crate's velocity just before landing is 32 m s^{-1} to prevent damaging the supplies.

(i) Show that the maximum height of point P is 52 m.

[1]

(ii) Point R is vertically below point P.

Calculate the horizontal distance QR.

QR = ... m [2]

(d) Suggest and explain the new landing point relative to Q if air resistance is not negligible.

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....

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.... [2]

[Total: 7]